

BitFlip

CloudSherpa

User Manual

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1 | Welcome to CloudSherpa

CloudSherpa is an intelligent cloud cost management platform designed to help organizations gain complete visibility into their cloud spending across multiple cloud providers. By combining cloud usage, costs and insights into a single, intuitive finOps dashboard, it eliminates the need to navigate between multiple cloud provider platforms.

The platform enables users to monitor cloud expenditure, identify unexpected cost increases and receive proactive recommendations to optimize resource usage and reduce unnecessary spending. Through real-time insights and predictive analytics, CloudSherpa helps organizations make informed financial decisions while maintaining efficient cloud operations.

By providing a unified view of multi-cloud environments and actionable cost optimization recommendations, CloudSherpa empowers businesses to control cloud expenses, maximise the values of their cloud investments and confidently manage their cloud platform.

Do I need to be a technical expert?

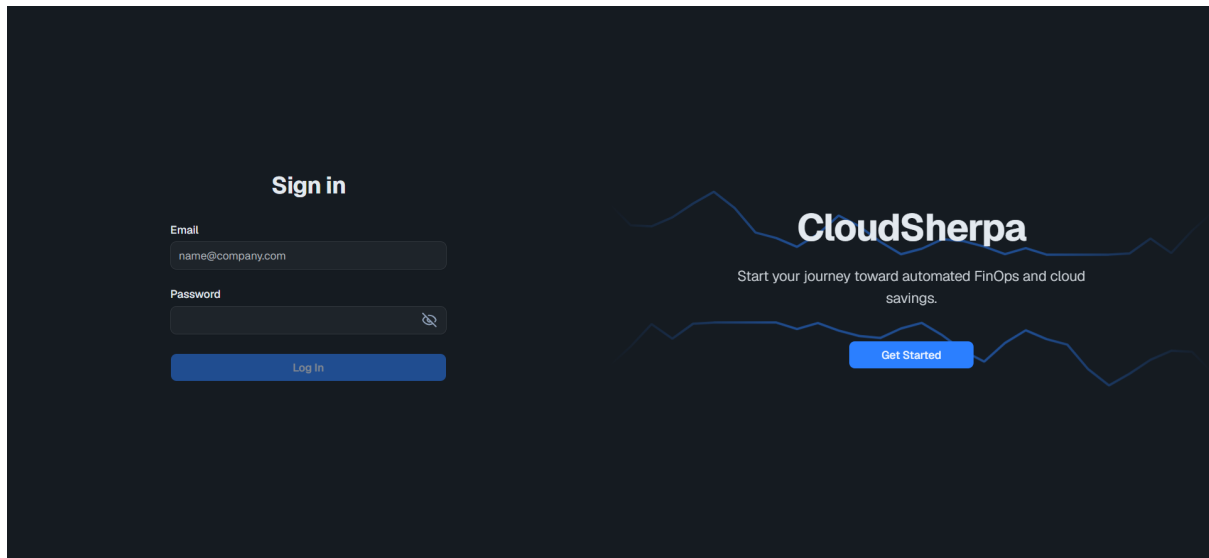
CloudSherpa is designed for both technical and non-technical users. It features a user-friendly dashboard that presents insights in a clear, intuitive manner, which enables users of any technical level to easily understand what is occurring.

2 | Getting Started

2.1 Accessing the System and Logging In

How to access the live deployment of CloudSherpa:

1. Navigate to <https://cloudsherpa.gjjcs.org/>



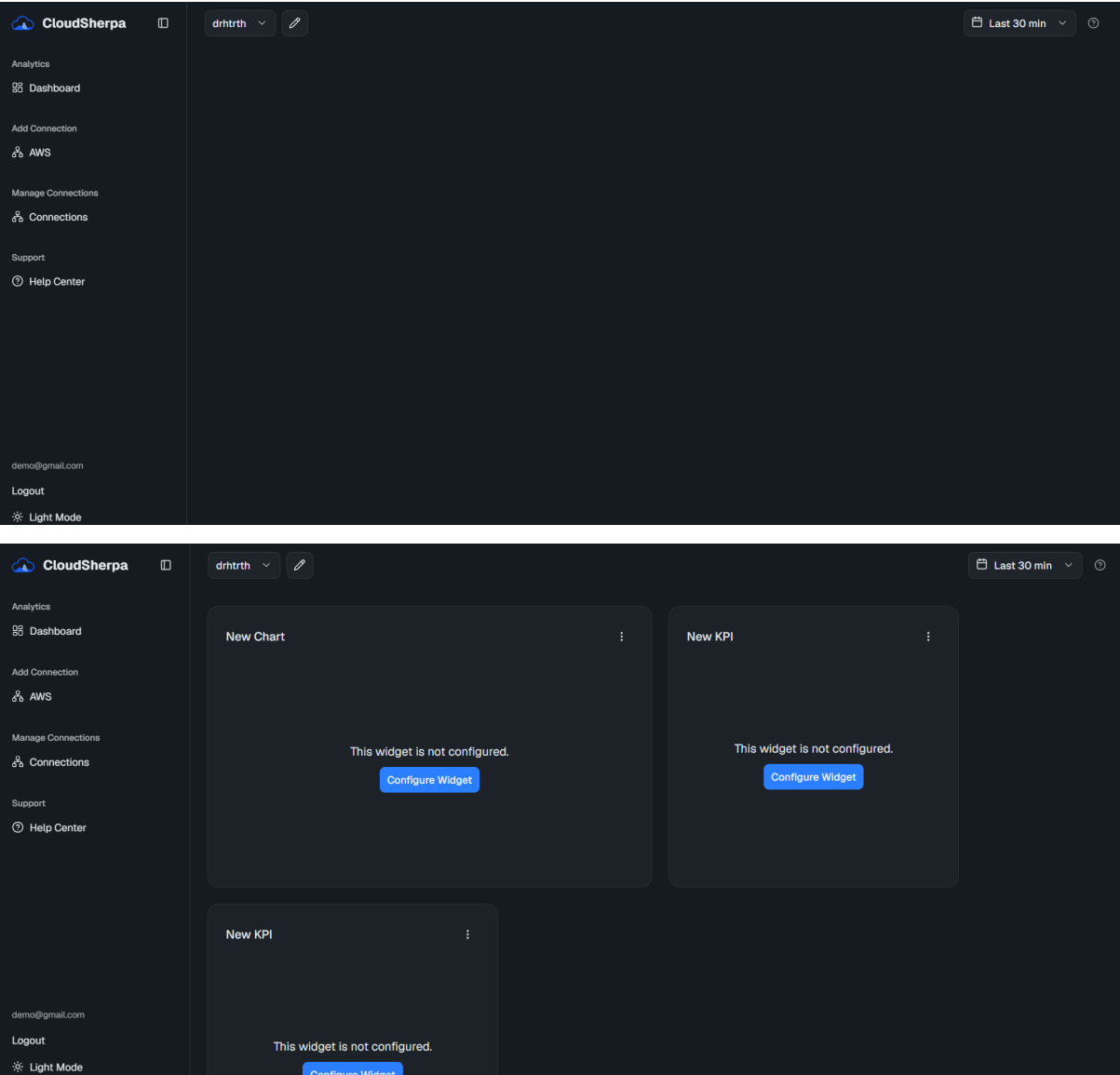
To create an account with CloudSherpa, follow these steps:

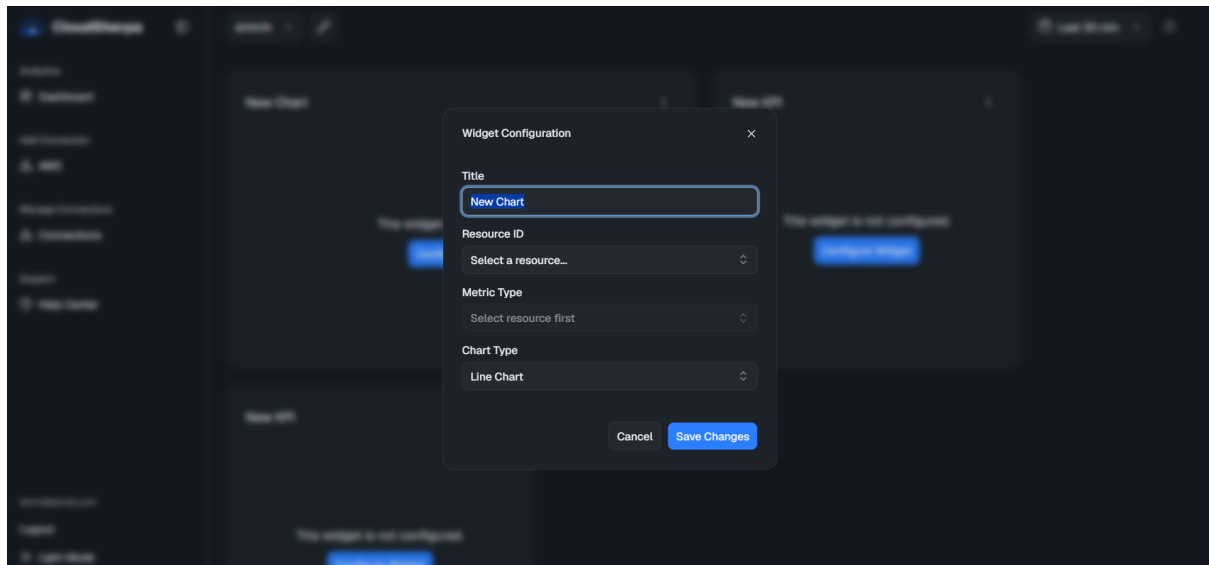
1. If you are a new user:
 - (a) Click the **Register** button.
 - (b) Fill in the required information.
 - (c) Create a secure password and confirm your registration.
2. If you already have an account:
 - (a) Click the **Login** button.
 - (b) Enter your registered email and password.
 - (c) Click **Login** to access your dashboard.

By following these steps, you can immediately begin using CloudSherpa's features.

2.2 The Main Dashboard

Upon logging in the user is directed to the dashboard:





Key areas of the dashboard:

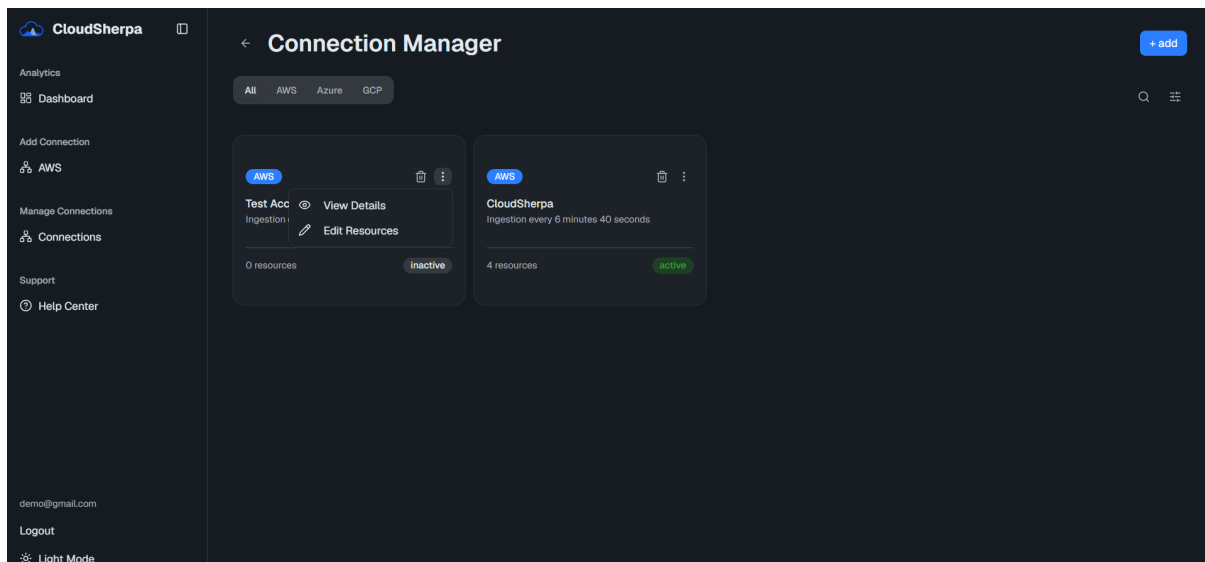
- **Navigation bar:** Users are able to navigate to various connections such as AWS. They are able to log out via the navigation bar or go back to the dashboard. The navigation bar is collapsible to give users a better view of what is being displayed on their screen.
- **Period selector:** The user can select which time period they would like to view the data being displayed. There are fixed period selectors such as 'last 7 days' or they can create a custom period that they desire.
- **Multiple dashboard:** More than one dashboard can be created via the dropdown. The users can also navigate to dashboards that were previously created via the dropdown.
- **Customize widgets:** Users can design the dashboard in a way they desire. The widgets size can be adjusted and their positions can be changed. This can be done by using the pen icon (next to the dropdown). They can further choose a graph they would like the data to be displayed in, from the options provided. The name of the widget can be customized and different types of data can be selected to be displayed eg.CPU.

3 | Core Features and Usage

These are the five core features of CloudSherpa

3.1 Feature 1: Connection Manager

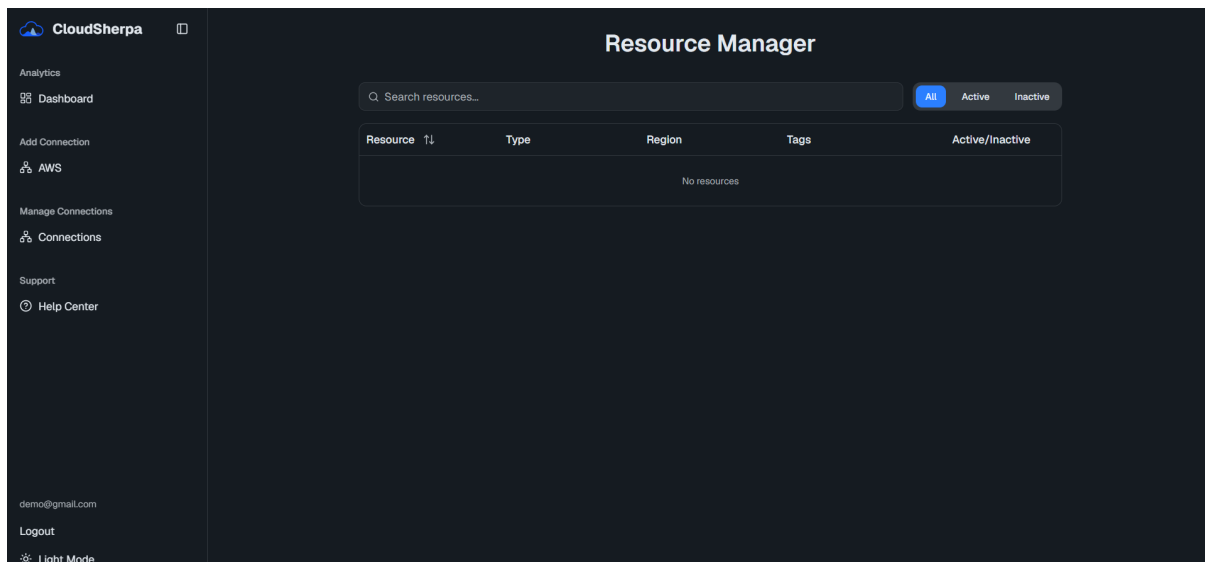
The connection manager is utilized to add multiple AWS connections where their services can be monitored as well as specific resources.



1. Locating connection manager - Navigate to the Connection manager (under Manage Connections) in the side navigation bar
2. Adding an AWS connection (under Add connection) - you will be taken through a setup wizard. In the first step you enter your credentials, the next step you choose which services you would like CloudSherpa to monitor and lastly you select which resources you would like add for that particular service.
3. Managing your connections - by clicking on the view details option. You can change the name of the connections and view the resources that is associated with that connection. Multiple connections can be added and deleted.

3.2 Feature 2: Resource Manager

The resource manager is used to manage your connections. You can make certain resources active and inactive. You can filter and search for a resource.



1. Locating the resource manager - In the connection manager configuration (was shown in previous feature), click on the edit resources button. You can successfully configure your resources here.
2. Active/Inactive resources - toggle the resources to determine which resource should be monitored for that particular service or not.
3. Step 3 Search and filter resources - Use the search bar to search for a resource or to filter click on the Resource name.

3.3 Feature 3: Scheduler

The scheduler scans the database every 30 seconds for the usage scheduler and 5 minutes for the billing scheduler.

We read in a field that informs us that an account is due and if they are they get enqueued to Jobrunr background jobs queue which is managed by Jobrunr.

Jobrunr spins threads in the background, up to 8 in our configuration. Each job attempts ingestion for only one account. If it fails, it retries 3 times and if it has still failed the appropriate field in the database will not get updated which mean that when the scheduler scans again that account will be scheduled again.

If it was successful, the appropriate next ingestion field gets updated in the database

such that the next scan will not pick the same job again

3.4 Feature 4: Setup wizard

The setup wizard consists of 3 steps, which each user has to go through each time they want to establish an AWS connection.

The screenshot shows the CloudSherpa web application interface. On the left is a dark sidebar with navigation links: Analytics, Dashboard, Add Connection, AWS, Manage Connections, Connections, Support, and Help Center. The main content area is dark gray and displays a modal window titled 'STEP 1 OF 3 Link your AWS account'. The modal contains instructions to log in to an AWS account and create a new IAM user. It features three input fields: 'Account name' (containing 'AWS Connection'), 'Access key ID' (containing 'EXAMPLE'), and 'Secret access key' (containing a series of dots). A blue 'Next' button is located at the bottom right of the modal. The browser's address bar shows the URL 'cloudsherpa-staging.gijcs.org/addConnection/aws'. The top of the browser window includes standard navigation icons and a VPN status indicator.

CloudSherpa

Analytics

Dashboard

Add Connection

AWS

Manage Connections

Connections

Support

Help Center

demo@gmail.com

Logout

Light Mode

cloudsherpa-staging.gijcs.org/addConnection/aws

☆

VPN

🔒

☰

STEP 2 OF 3

Configure Billing & Services

☒ Opt-in to billing

BILLING EXPORT CONFIGURATION

Account-wide cost scope

S3 Bucket Name

e.g., my-billing-reports-bucket

Bucket region

Select a region

Export Name

e.g., daily-cost-export

Prefix / Path

e.g., cur-reports/2023/

Save Billing Config

SERVICES FOR USAGE MONITORING & RESOURCE DISCOVERY

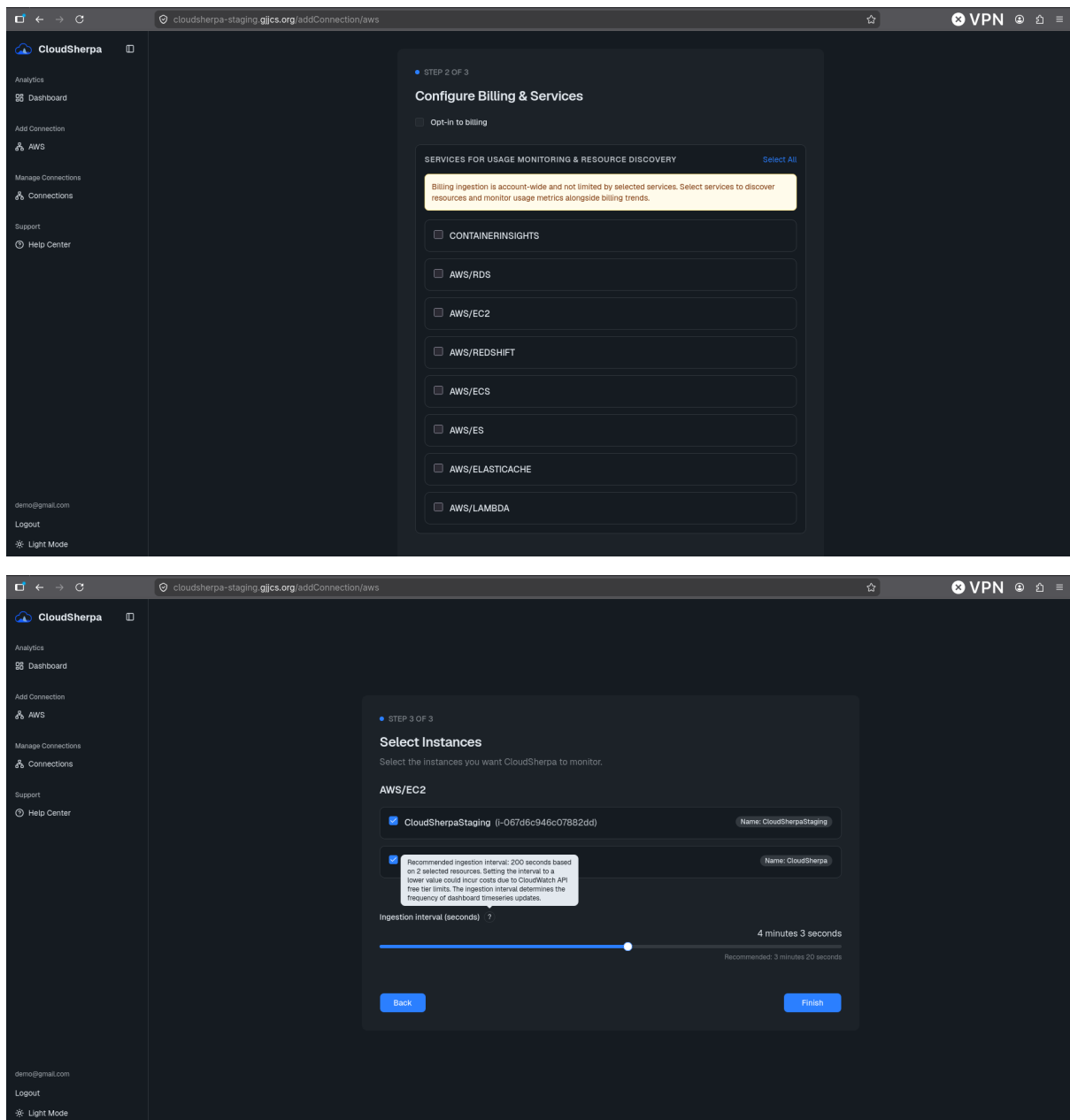
Select All

Billing ingestion is account-wide and not limited by selected services. Select services to discover resources and monitor usage metrics alongside billing trends.

☐ CONTAINERINSIGHTS

☐ AWS/RDS

☐ AWS/EC2

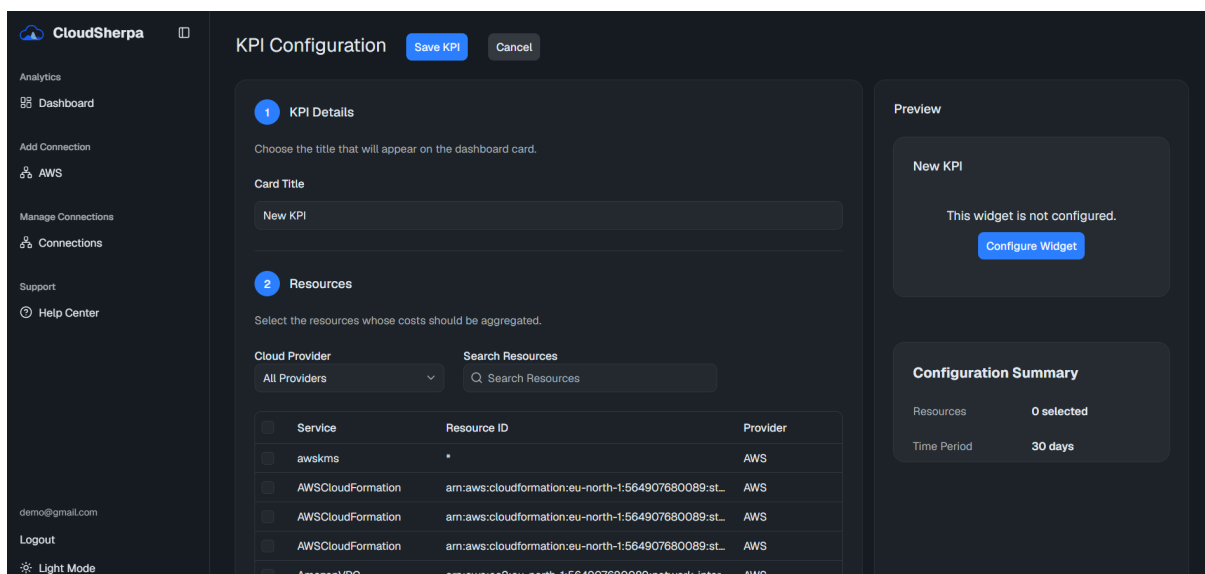
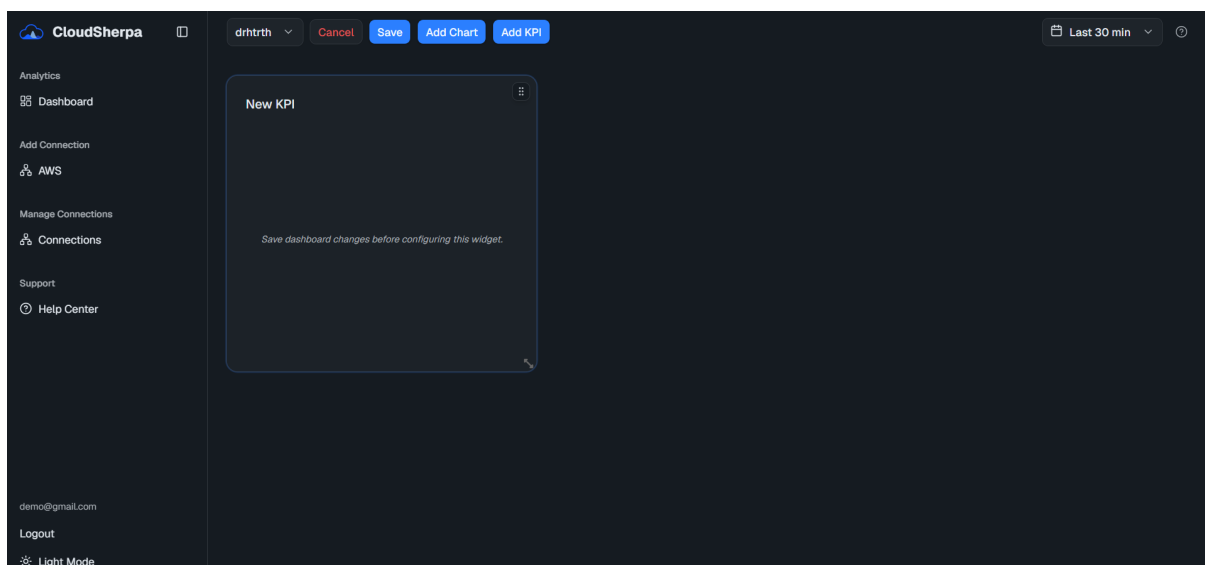


1. Step 1 - Provide the credentials for IAM user. You can choose what you would like to call the connection. (can be changed at any time under Connections)
2. Step 2 - Select the services which you would like CloudSherpa to monitor. The billing in this step is optional and you can also opt in or out of usage monitoring for specific services.
3. Step 3 - Displays the resources that are associated with the services that were in the previous step. You can select which ones you would like CloudSherpa to

monitor. (can be manager anytime under Connections)

3.5 Feature 5: KPI widget, configuration and billing ingestion

It allows the user to view all of their cloud charges that went off against their account. On a granular level they can select which charges to monitor or omit and at a time period that they prefer. This way the user can build a custom view of their cloud spending and gain useful operational insights to make informed decisions.



The screenshot displays the CloudSherpa dashboard interface. On the left is a sidebar with navigation links: Analytics, Dashboard, Add Connection, AWS, Manage Connections, Connections, Support, and Help Center. The main area features a table with columns for Service, Resource ID, and Provider. The table lists various AWS services like AWSKMS, AWS CloudFormation, Amazon VPC, Amazon ECR, AWS Data Transfer, and Amazon ECS. Below the table, there's a pagination bar showing '0 of 35 row(s) selected', 'Rows per page' set to 10, and 'Page 1 of 4'. A 'Time Period' widget is visible on the right, allowing users to select a time window for cost aggregation, with '30 days' currently selected. The bottom left of the dashboard shows the user's email 'demo@gmail.com', a 'Logout' button, and a 'Light Mode' toggle.

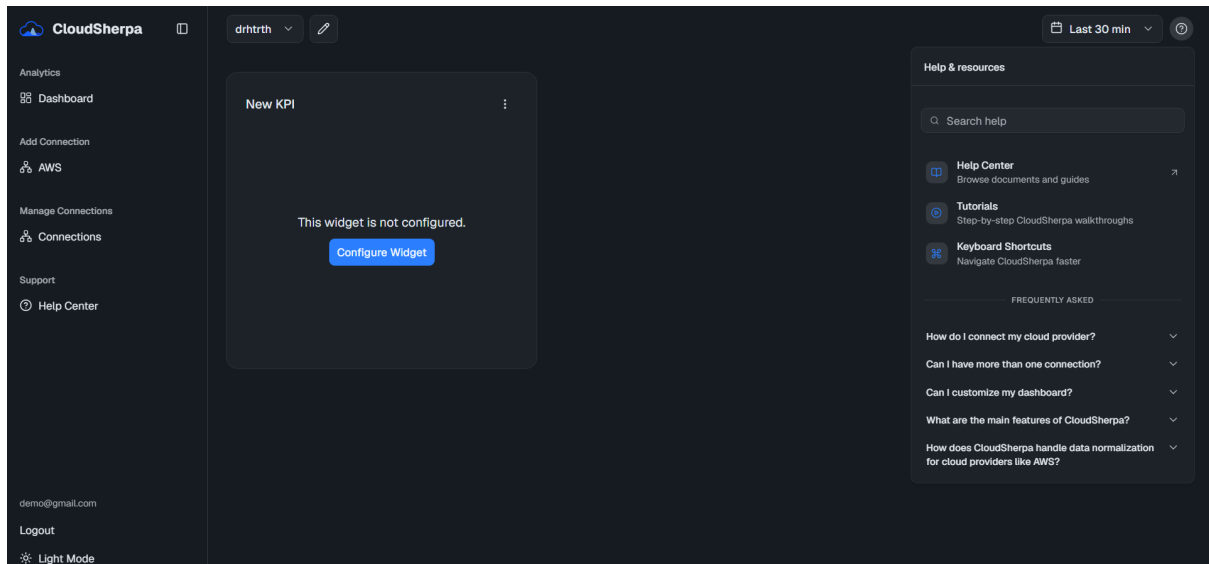
Service	Resource ID	Provider
aws kms	*	AWS
AWS CloudFormation	arn:aws:cloudformation:eu-north-1:564907680089:st...	AWS
AWS CloudFormation	arn:aws:cloudformation:eu-north-1:564907680089:st...	AWS
AWS CloudFormation	arn:aws:cloudformation:eu-north-1:564907680089:st...	AWS
Amazon VPC	arn:aws:ec2:eu-north-1:564907680089:network-inter...	AWS
Amazon VPC	arn:aws:ec2:eu-north-1:564907680089:network-inter...	AWS
Amazon VPC	arn:aws:ec2:eu-north-1:564907680089:network-inter...	AWS
Amazon ECR	arn:aws:ecr:eu-north-1:564907680089:repository/clo...	AWS
AWS Data Transfer	arn:aws:ecs:eu-north-1:564907680089:task/cloudshe...	AWS
Amazon ECS	arn:aws:ecs:eu-north-1:564907680089:task/cloudshe...	AWS

1. The user can change the name of the KPI widget.
2. They can select the specific billing records they would like to monitor.
3. Users can also select which time window they prefer for the billing.

4 | In-System Help Menu

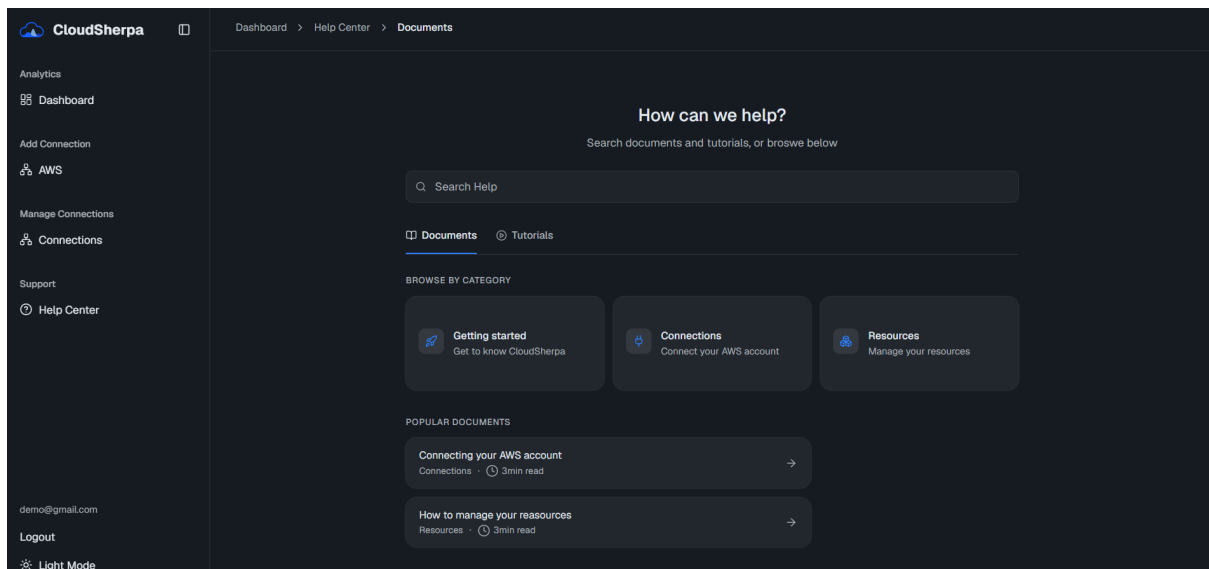
To allow for easy navigation around CloudSherpa, we have included a help menu which users can utilize to familiarize themselves with the features of the platform before delving into it.

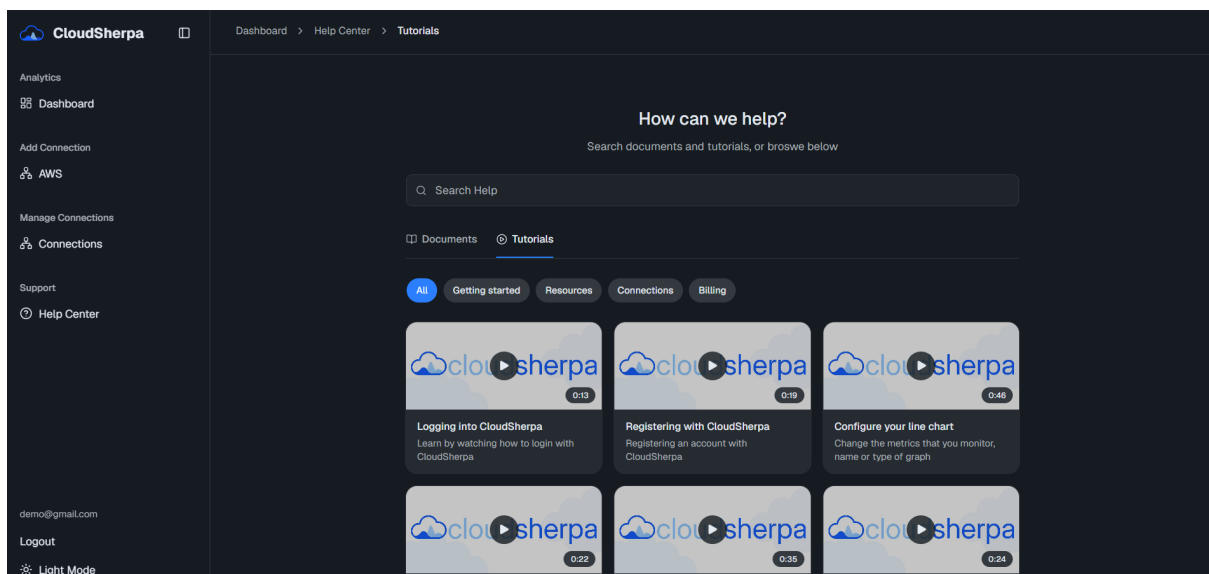
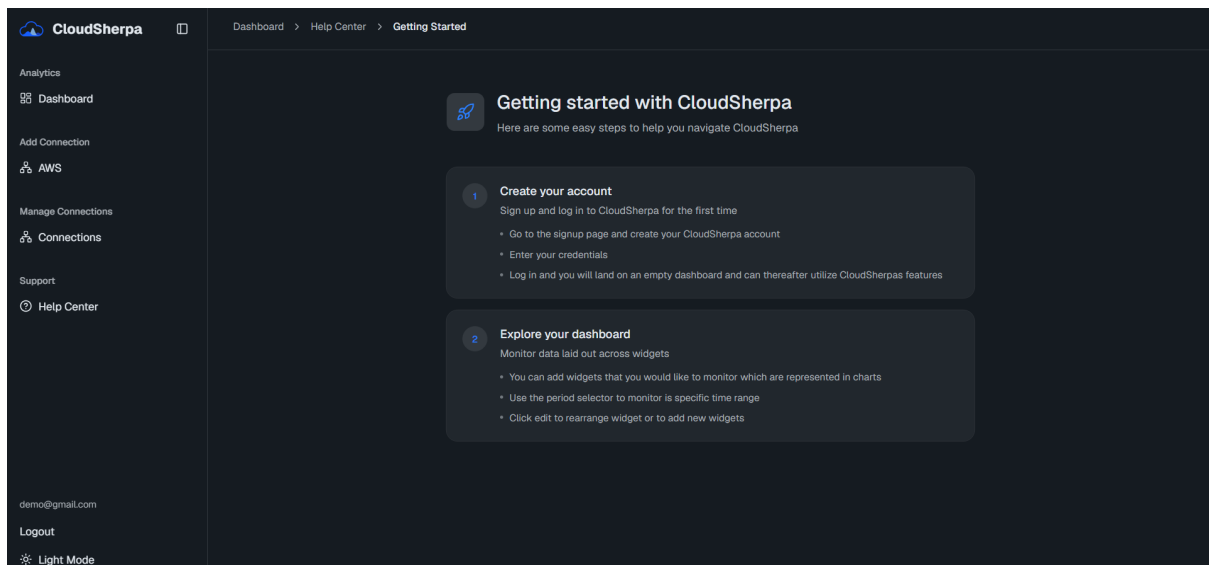
4.1 Accessing Support



The help menu can be accessed from the top-right hand corner of the dashboard by clicking on the ? or in the side navigation bar under support.

When clicked upon, the user is provided with a list of options such as: Help center, tutorials, Keyboard shortcuts and frequently asked questions.





Help center provides detailed documentation and tutorial videos which can help a user to understand how the features of CloudSherpa work. The documents provide step by step instructions on how to execute a specific task. It is also accompanied by tutorial videos that demonstrate how the instruction can be executed.